

A Two-Color Photometric Method for Detection of Extra-Solar Planetary Systems

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The transit of a star by a dark companion or planet results in a characteristic colorimetric "signature," which should be detectable by photometric means. This signature, which is due to differential limb darkening in red and blue light, takes the form of a slight shift towards the blue as the planet crosses the limb of the star, followed by an abnormal reddening during the transit of the center region of the star, and finally another shift towards the blue as the planet approaches the far limb. An analysis of this signature, the probabilities of detecting such transits (using the solar system as a model), and possible instrumentation for this purpose are discussed. A system designed to yield one or more planetary detections per year is believed to be feasible at moderate cost. Such a system would consist of three wide-field telescopes, at well-separated sites, slaved to a central computer. Each telescope would be equipped with a two-color photometer employing an image dissector under computer control for tracking and rapid switching between program stars without requiring frequent resetting of the instrument. Each telescope would survey an independent sample of stars on a rapid rotation cycle, until one of the telescopes detected an unusual transient, at which time all three would be used to provide additional data and increased reliability as a coincidence detection system.

I. INTRODUCTION

This paper proposes a means for detecting dark companions and planetary bodies by photometric observations of stellar eclipses. Unlike previously discussed techniques, this method relies not on the absolute change in luminosity which occurs in an eclipse, but rather on a characteristic colorimetric "signature" which can be measured as a change in a ratio rather than in absolute intensity. It is proposed that the atmospheric noise limitations on conventional photometric work can be further reduced by employing a number of widely separated small telescopes under the control of a central computer, and employing an image dissector tube at each telescope to permit virtually simultaneous comparison of a number of stars without resetting the telescope. By the combination of star-to-star comparison, forward prediction, and coincidence detection techniques, as well as the use of measurements

in one color to predict changes in another, it appears that at least one and possibly two orders of magnitude can be gained over the accuracy of more conventional techniques for the measurement of transient phenomena.

In addition to the planet detection problem, this would seem to put a number of other interesting problems within reach. These include the measurement of stellar rotation by means of autocorrelation of fluctuations attributable to sunspots; the observation of short-term instabilities such as stellar flares; the study of light curves of small amplitude variables; and the determination of the photometric noise spectra of different stellar types. Since most of these applications are straightforward, however, given the basic capabilities of the technique, we shall limit our attention in the following to the detection of dark companions and planetary systems, which does envisage some procedures which are less self-evident.

The incidence of planets in the galaxy remains, at this time, a matter of speculation. A plausible model, advocated by Su-Shu Huang (1963), suggests that the reduction in angular momentum which occurs in main-sequence stars of spectral type F5 and later types is associated with the formation of planetary systems.

Another approach is to extrapolate the statistics of multiple-star systems (noting that the majority of stars are binary) down to the level of planetary size. This generally yields a more conservative (but possibly less accurate) estimate. A related technique is to examine the closest known stars for invisible companions of small mass by means of motion perturbations. This has revealed the presence of bodies approaching planetary size in several cases. These include Strand's discovery of a companion of 61 Cygni having 0.01 solar mass, and a body having 1.5 times the mass of Jupiter orbiting around Barnard's star. A re-evaluation of this last example by Van de Kamp (1969) suggests that it may be more correctly interpreted as a pair of planets orbiting at 2.8 and 4.7 A.U. with masses of 0.8 and 1.1 times the mass of Jupiter respectively. The discovery of two planetary systems among the very few stars close enough to observe motion perturbations suggests that such systems may indeed be common, but these two cases are hardly adequate to make any inferences about the statistics of planetary systems, sizes and numbers of planets, association with different stellar types, or other questions of fundamental interest.

Both Struve (1952) and Huang (1963) have examined the feasibility of planetary detection using astrometric, spectroscopic, and photometric means, and have concluded that the prospects of detection are remote with available techniques. A sun-type star at a distance of ten parsecs, and a planet comparable to Jupiter, for example, would require the measurement of an angular deviation of $0.0005''$, a radial velocity measurement accurate to 0.01 km/sec, or (to detect an eclipse by absolute measurements) the detection of a luminosity change of 0.01 magnitudes. This last measurement is the only one which is

considered to be feasible, but it is noted that the probability of observing an eclipse of a planet in Jupiter's orbit, seen from a random direction and at a random time, is negligibly small, so this method has also been dismissed as impractical.

In the following sections, we shall reconsider the eclipse technique. First of all, let us examine the probability of detecting an eclipse with a hypothetical perfect detector, taking the solar system as a model, and without introducing the arbitrary assumption that we will consider only planets in a Jovian orbit, or of giant size. We will then examine, as a separate problem, the matter of devising a suitable technique for detecting such eclipses. From this treatment it will be shown that the probabilities are by no means as unfavorable as has been assumed, and that practical systems that can accomplish such detections are indeed worth considering. In fact, the feasibility of such a program, with telescopes of relatively small size and cost, makes this seem particularly inviting as an application for the large number of university based instruments which might readily be adapted to such work. It should be emphasized, of course, that any results obtained in this fashion are to be taken merely as evidence supporting a hypothesis, until repeated observations (perhaps with the collaboration of space-based telescopes for final confirmation) have established the periodicity and details of the light curve beyond doubt. What is proposed here is a survey and detection procedure, with final confirmation depending upon collaborative work by a number of instruments over an extended time period.

II. THE PROBABILITY OF STELLAR ECLIPSES

Consider our solar system from the point of view of an observer situated at infinite distance, in a randomly chosen direction. What is the probability that within a 6-hour period of observation (corresponding to a typical night's work) he would observe either the onset or the termination of an eclipse by any one of the sun's planets? For the moment, we will assume that he can recognize the transient

changes associated with onset or termination regardless of the magnitude of the change.

For the i th planet, the probability of observing an eclipse, P_{obs} , subject to these assumptions, is the product of two factors:

$P(t_i)$ = probability of onset or termination of an eclipse occurring within a given 6-hour period;

$P(\theta_i)$ = probability that the orbital path of the i th planet passes through the cone from the observer to the perimeter of the sun.

For the first of these factors, we have

$$P(t_i) = 6 \text{ hours/orbital period} \times 2. \quad (1)$$

For the second, note that with the observer stationed at infinity, the required quantity is the probability that the line of sight falls within an angular distance $\theta_i/2$ from the orbital plane, where θ_i is the angle subtended by the sun from the distance of the i th planet. This gives

$$P(\theta_i) = 2\pi\theta_i/\frac{4}{3}\pi = \frac{3}{2}\theta_i, \quad (2)$$

where

$$\theta_i = 2 \times \arctan 432\,000/D_i,$$

D_i = distance of i th planet from the sun, in miles.

A further quantity of interest is τ_i , the transient time for the i th planet to cross the limb of the sun, during an eclipse. Again, as seen from infinite distance, for an eclipse near the solar equator this is equal to the diameter of the planet divided by its orbital velocity.

Table I lists these quantities, as well as other pertinent data for the planets of the solar system.

It is worth noting that for the more remote planets, the duration of an eclipse is so prolonged that the probability of observing an eclipse in progress, in any given 6-hour period, exceeds the probability of observing a beginning or termination. Since we shall be concerned with the "signature" of a planet crossing the limb of the sun, however, it is the probability of observing this transient event which is

computed for the column headed P_{obs} . It should be noted that the duration of this transient (τ_i) ranges from less than two minutes up to several hours. For increasingly eccentric eclipses, the transient time is prolonged; for example, for an eclipse crossing the solar disc at 60° the time should be roughly doubled.

If we now turn our attention to the joint probability of observing an eclipse of at least one planet during a given 6-hour period, this is closely approximated by the sum of P_{obs} . This is approximately 0.0003, the main contribution coming from the inner planets. In other words, assuming perfect detection capability, about one 6-hour observation in 3000 would be expected to reveal the beginning or termination of an eclipse. We should note immediately, however, that rapid scanning or simultaneous observation of a set of, say, ten stars of an appropriate planet-bearing type would reduce the number of observing nights per detection by an order of magnitude, giving an expected detection rate of about one system per year. About one star in a hundred would have its orbital plane close enough to the line of sight so that all four inner planets could be detected.

All of this assumes planetary systems identical to our solar system. What are the probable effects of variations from this model?

For stars of a constant mass, it is easy to see that P_{obs} increases proportionally with the stellar diameter, for a planet at a given distance from the primary. Also, for stars of constant diameter, P_{obs} increases proportionally to the square root of the mass of the star (due to orbital acceleration). For a star of uniform density, P_{obs} increases with the $5/2$ power of the diameter due to the combined effect of orbital acceleration and greater admissible viewing angle. Thus doubling the diameter of a star (of the same average density) would result in a 5.6-fold gain in P_{obs} , for a planet at a given distance. For a typical star, however, the mass increases roughly with the square of the radius; consequently, we should expect P_{obs} to increase with the square of the stellar radius under typical

conditions. Since some of the stars of interest are appreciably larger than the sun, this suggests that the probability estimates given in Table I may be conservative in an appreciable number of cases. For the majority of observations, however, stars of solar size or less seem to be the most likely candidates.

III. ECLIPSE SIGNATURES

So far we have assumed an ideal ability to detect eclipses, no matter how small the diameter of the planet relative to that of the star. In the case of Mercury, this would mean a reduction of only 10^{-5} of the light of the sun; in the case of the Earth or Venus about 10^{-4} , and for the larger planets 10^{-3} to 10^{-2} . While there would be no difficulty in making such measurements for the large planets, we have seen that it is the inner (and smaller) planets which make the principal contribution to P_{obs} . Conventional photometric work is never accurate to better than 1% (0.01 magnitudes) and rarely precise to a few thousandths (e.g., see McCullough, 1964). We must now turn, therefore, to a con-

sideration of techniques by which eclipses can be detected more reliably than by measurement of changes in the total light flux.

The method which is proposed here makes use of the differential limb-darkening of a star in the blue and red portions of the spectrum. A scan line across the sun in the infrared is virtually flat, showing the same intensity at the limb as at the center. As we move towards the ultraviolet, however, we find that the intensity at the limb is only a small fraction of that at the center. Figure 1, based on data from Zirin (1966) shows intensity curves as a percent of center intensity for wavelengths from 3200 to 12 000 Å. A table showing expected limb darkening for different types (based on Kopal, 1959) is included as Table II. Observed darkening (cf. Wood, 1963) is typically somewhat less due to blanketing effects.

Assume that the ratio of a star's light in the IR and UV is monitored continuously, as an eclipse (or stellar transit) occurs. At the beginning of the eclipse, when the planet just crosses the edge of the star,

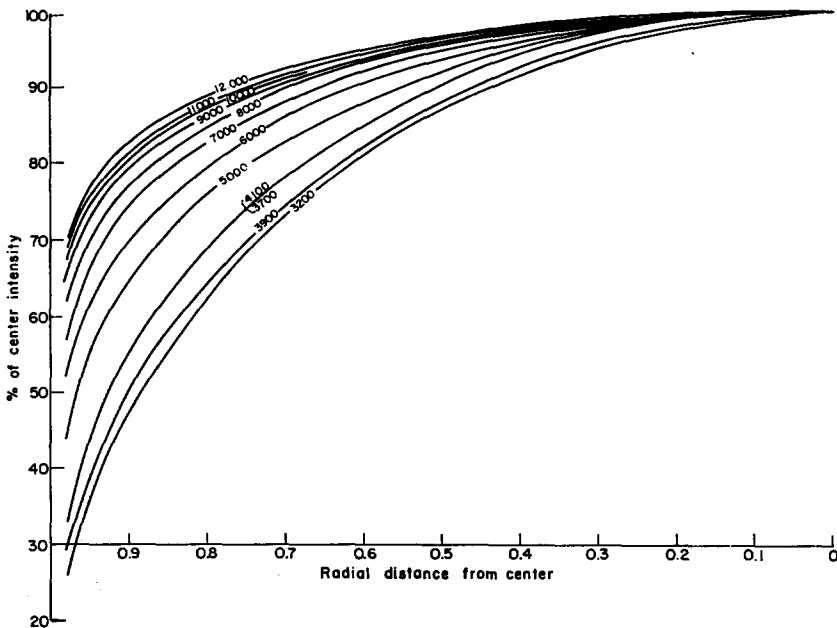


FIG. 1. Solar limb darkening curves from 3200 to 12 000 Å.

TABLE II
MONOCHROMATIC COEFFICIENTS u_1 OF THE LINEARIZED LAW OF LIMB-DARKENING
FOR MAIN-SEQUENCE STARS OF DIFFERENT SPECTRAL TYPES^{a,b}

Spectrum	3647-	3647+	4000	Effective wavelength (Å)		6500	8203-	8203+
				4600	5300			
B0	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
B5	0.50	0.73	0.70	0.65	0.62	0.55	0.49	0.50
A0	0.38	0.79	0.75	0.67	0.60	0.49	0.37	0.39
A2	0.34	0.80	0.77	0.68	0.61	0.48	0.34	0.38
A5	0.41	0.82	0.79	0.70	0.62	0.50	0.35	0.38
A7	0.48	0.83	0.80	0.71	0.64	0.51	0.37	0.40
F0	0.58	0.85	0.82	0.73	0.65	0.53	0.39	0.43
F2	0.64	0.86	0.83	0.74	0.66	0.54	0.41	0.45
F4	0.71	0.87	0.84	0.76	0.68	0.55	0.42	0.47
F6	0.78	0.89	0.84	0.77	0.69	0.57	0.44	0.49
F8	0.84	0.90	0.85	0.78	0.70	0.58	0.46	0.51
G0	0.89	0.92	0.86	0.80	0.72	0.59	0.47	0.53
G2	0.92	0.93	0.87	0.81	0.74	0.61	0.49	0.55
G5	0.94	0.94	0.88	0.83	0.76	0.64	0.52	0.57
K0	0.96	0.95	0.91	0.86	0.79	0.67	0.56	0.62

^a Based on linear model of Kopal (1959).

^b If γ is the angle between the radius vector and line of sight, the relative intensity for a given frequency (by linear law of limb darkening) is equal to $1 - u + u \cos \gamma$.

it is obscuring proportionally more of the red light than of the blue, resulting in a transient diminution of the red/blue ratio. As the planet moves towards the center, this effect is reversed, with the planet at the center obscuring proportionately more blue light than red, so that the R/B ratio rises above normal. Finally, as the planet approaches the far limb, the ratio goes below normal again, to return abruptly to its normal level as the eclipse terminates. For an eclipse which crosses the stellar disc far from the equator, the negative phases are prolonged, and for a sufficiently eccentric eclipse the positive phase does not occur.

More specifically, the equations characterizing the observed light diminution due to an eclipse at wavelength λ are as follows. The mean intensity over the stellar disc is

$$L(\lambda) = 2\pi \int_0^r I_{r,\lambda} r dr, \quad (3)$$

where $I_{r,\lambda}$ = intensity at radius r and wavelength λ . The proportional eclipse at r,λ (i.e., the fraction of light lost at wave-

length λ due to an obscuring planet at radius r) is

$$\Delta_{r,\lambda} = \frac{I_{r,\lambda}}{L(\lambda)} \cdot a \quad (4)$$

where a is the fraction of the stellar disc eclipsed by the planet. The colorimetric "signature" of the eclipse, for wavelengths λ_1 and λ_2 , is given by

$$E_r(\lambda_1, \lambda_2) = \Delta_{r,\lambda_1} - \Delta_{r,\lambda_2}. \quad (5)$$

Solutions of Eq. (4) for various values of λ , based on Zirin's data for the sun, are shown in Fig. 2, where it is assumed that the area eclipsed (a) is 1% of the solar disk. The resulting eclipse signature, measured at 9000 and 3200 Å, is shown in Fig. 3, for eclipses crossing the stellar disc on three different paths. For an equatorial eclipse, the difference between minimum and maximum red/blue ratio is 0.7%. Note that for any planetary transit of the sun (where the duration of an equatorial eclipse ranges from 8 hours for Mercury to 62 hours for Neptune) the negative phase can be expected to last for at least an hour, the initial downward transient

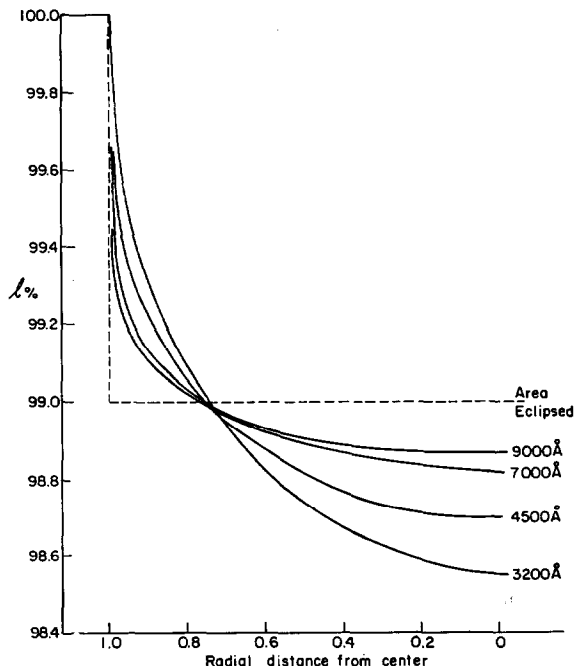


FIG. 2. Eclipse due to 1% area coverage, as function of radial distance and wavelength.

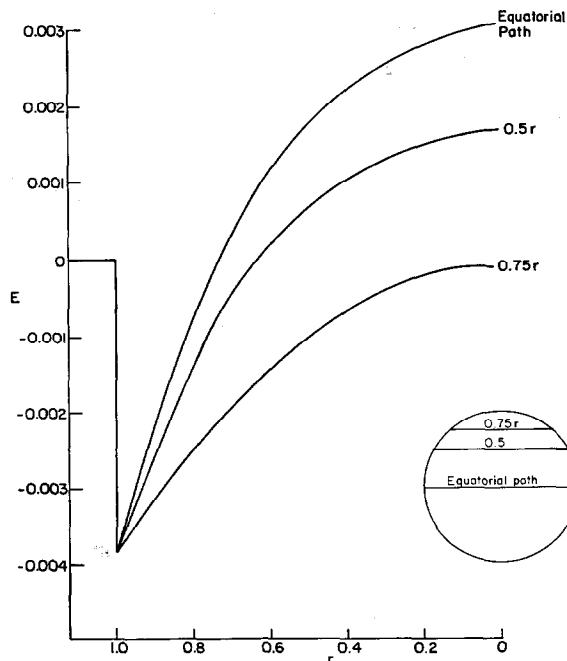


FIG. 3. Signature curves for eclipse of 1% of solar area. $\lambda_1 = 9000 \text{ Å}$; $\lambda_2 = 3200 \text{ Å}$.

taking from 103 seconds for Mercury to over three hours for Saturn.

Since this type of signature involves a differential measurement (or a measurement of a ratio) rather than an absolute measurement as in the determination of a stellar magnitude, it is likely that it can be made with considerably greater precision. The extent to which this is true will depend, of course, on the types of noise effects and possible artifacts which might be confused with an eclipse. In the following section, we will examine the factors most likely to contribute to errors in detection and measurement, and will then turn to the techniques of analysis which seem most effective for minimizing such errors.

IV. NOISE SOURCES

Three main sources of "noise" (including any artifacts or spurious signals which might be confused with those being sought) can be distinguished. These are: (1) Stellar noise, due to intrinsic variation in the light flux from the star under observation, or from comparison stars which might be used as a standard; (2) Atmospheric noise, due to inhomogeneity and variations in transmission by the Earth's atmosphere; (3) Instrumental noise, including variations in photodetectors, electronic circuitry, recording errors, etc. The first of these three types of noise is clearly inescapable, and if it is of sufficient magnitude, it sets limits on the use of photometric techniques for planetary detection which we can do nothing about. The other two sources can each be reduced by various techniques which will be considered in Section V. In this section, we will try to estimate the magnitude of each of the noise components.

A. Stellar Noise

Unfortunately for purposes of planetary detection, we cannot expect the observed star to be a perfectly stable and uniform emitter of light. The main sources of stellar light variation are expected to be the following:

1. *Intrinsic stellar variation:* Known

variable stars are likely to be poor candidates for a study of this type. Large amplitude variables, particularly those with a regular period, can be readily distinguished. Eruptive variables, and stars having low-amplitude irregular light curves require more careful consideration, as possible sources of artifacts. Discrimination of such effects from eclipse effects should be possible, however, by making use of the following points: (a) The time course of the signal is apt to be quite different. While the planetary eclipse signal has a perfectly symmetric form, with an abrupt onset and termination typically taking only a few hundred seconds, intrinsic stellar variations tend to be asymmetric and to develop and fade out over a much longer time period. With the exception of R. Coronae Borealis variables (Glasby, 1969) the variation always represents an increase of luminosity over the normal phase, while an eclipse causes a decrease. (b) The colorimetric signature and transient spectral variation are apt to be quite different. While a planetary eclipse is characterized by a transient loss of red, followed by an abnormal reddening (the middle phase), and then another reduction of the red component, a typical variable star curve will show spectral changes which develop to a maximum and then diminish, rather than showing the typical three-phase eclipse pattern. (c) The finding of a signature should, in any case, serve simply as a strong indicator of the presence of a planetary system; confirmation depends upon repeated detection, with a determination of the period and orbital characteristics. An inability to obtain such confirmation would lead to a rejection of the finding.

The probability of eclipse-like artifacts being generated by virtue of low amplitude stellar variability is hard to assess without observational data at the level of sensitivity that we are considering (luminosity variations of 10^{-4} or less). Observations of the sun are most unsatisfactory as a baseline, since they involve atmospheric effects, and require integration over the solar disc from bolometric systems which rarely have had the precision which we require. The best

measurements of stability of solar radiation, in fact, have been obtained by the indirect method of measuring variations in the reflection of Neptune and Uranus in comparison to standard stars, a method quite similar to that which we are proposing (Johnson and Iriarte, 1959). Long-term variations of the solar constant, of up to 2% over 6 years, have been suggested by some studies, but have not been demonstrated, and skepticism is in order. In any case, this involves a time-constant which need scarcely trouble us. How much intrinsic short-term variability exists in stars of different types is, in fact, one question of interest which might be answered by a preliminary survey with the proposed system.

2. *Variation due to companion stars:* Multiple stars, where the companion is massive enough to produce distortions in the shape of the primary, or where reflection varies with position, may lead to variations of apparent luminosity with the motion of the observed system. Known multiple systems are probably poor candidates for a planetary search program in any event, due to orbital complexity and other irregularities. Smaller variations, due to companions approaching planetary mass, would be expected to have time constants far longer than those of an eclipse, and should not cause difficulty. Eclipsing binaries represent, of course, one end of the continuum which we are proposing to observe, and should be easy to recognize.

3. *Flares:* Flares have been observed on a number of red dwarf stars in addition to the sun (Glasby, 1969). In the sun, their magnitude (about 10^{28} ergs/sec (Brandt, 1966) may be as great as 0.25×10^{-5} of total solar luminosity, but in the UV Ceti variables, much greater variations are observed, frequently exceeding one magnitude. Their time-course, however, is typically a rapid increase (up to 0.05 magnitudes/sec) followed by a slow decay. These flares, which are strongest in blue light, would result in a color shift rather similar to that at the onset of a planetary transit. Known flare stars are mainly of spectral type M, which could be omitted in

a preliminary survey, but final discrimination would again rest on demonstration of periodicity of the suspected eclipse.

4. *Prominences and coronal effects:* Dramatic as prominences appear to be in photographs of an eclipsed sun, they represent, in fact, a negligible variation in the total light flux, hardly radiating at all in the continuum (Zirin, 1966). The entire corona is estimated to contribute less than 10^{-6} of the total solar radiation, and coronal variations are clearly below the level that we are concerned with.

5. *Sunspots:* Variations due to sunspots may be due to two causes: first, the appearance of a spot at the limb of a star, due to stellar rotation, might be expected to mimic the effect of a planetary eclipse; second, the appearance and disappearance of sunspots, or changes in shape and size, will contribute a noise component which might hide a planetary signature. The first of these effects is, fortunately, of a time scale much longer than the transient phase of an eclipse. The period of solar rotation is 27 days, and a sunspot first appearing at the limb is seen in profile, requiring a number of days to approach its maximum apparent area. Since the stars of interest are precisely those whose angular momentum is low, a similar period of variation might be expected in stellar observations. By contrast, the light-curve from an eclipse falls at once to about 66% of its full depth (depending upon the magnitude of limb-darkening), and completes its course in a small fraction of the period of solar rotation.

The birth and death of small sunspots, and changes in size of larger ones, are less likely to generate signals which systematically resemble an eclipse than they are to limit accuracy of observation by virtue of a statistical noise component. If we assume that most stars are characterized by "quiet" and "active" periods, as is the sun, then this noise could be minimized by limiting our observations to those stars which appear to be in a quiet phase. Observations of changes in solar fine-structure (Bray and Loughhead, 1965) show that changes in a small sunspot may proceed at a rate of up to 10 square seconds

of arc over a 10-minute period, or about 0.25×10^{-5} of the solar area. If we assume (as a rather severe case) that there are 100 such spots varying independently, this would give an rms variation of 2.5×10^{-5} over a 10-minute period, an effect roughly comparable to an eclipse of the sun by Mars. Since this is, in any case, pushing the limits of detectability, it is unlikely that sunspot variation would interfere excessively with the detection of larger planetary bodies. As in the case of all other possible artifacts, however, the ultimate check consists of determining the orbital period of the observed planet, and predicting its reappearance.

6. *Fine-structure variation in photosphere:* The main elements of solar fine-structure include bright spots, or plages, and granules. Sawyer (1968) estimates the maximum brightening from a single plage, in the visible range, as about 10^{-3} of the total visible radiation. The growth of such a plage requires a day or longer, however, and a longer period is required for decay. This gives a possible transient of about 10^{-4} per hour, as compared to at least an order of magnitude greater rate for the brightening at the end of an eclipse, with which this might otherwise be confused.

Granules, which are seen in high-resolution photographs, are estimated by Zirin (1966) to show a temperature variation of about 100°K , corresponding to about an 8% variation in luminosity between brightest and darkest points. There are on the order of a million such granules over the solar surface. The lifetime is variously estimated as five to ten minutes, but Bray and Loughhead (1965) demonstrate that there may be considerable stability over a 10-minute period. If we assume (contrary to observation) that granules form completely independently of one another, this would suggest an rms variation of about 10^3 granules, corresponding to a variation in total light of about 10^{-4} , with a time constant of about 10 minutes. Actually, however, formation of new granules does not occur independently, the total number per unit area remaining roughly constant, with new granules appearing in previously dark

regions as old ones disappear. Since this correlation between neighboring points tends to be negative rather than positive, the actual total variation due to this noise source is more likely to be on the order of 10^{-5} , or slightly larger. This would not interfere with observations except in the case of eclipses by the smallest planets.

7. *Conclusions:* While there are clearly a number of types of stellar variation, including variable star effects, sunspots, flares, and plages which are large enough to compare with light changes due to planetary eclipses, these phenomena can all be distinguished by different signs of the luminosity change, differences in spectral characteristics, different time constants, lack of symmetry, or some combination of these features. Background noise is due primarily to variations in the fine-structure and spot configurations of the stellar surface, and is unlikely to exceed 10^{-4} of the total luminosity, over a period of a quarter hour or longer. This means that planets down to earth size should be distinguishable from the noise background, while smaller planets could be detected marginally, at best, regardless of technical capability of the detecting system. In the case of the solar system, we would expect that all but two planets would be detectable, Mars probably being detectable under good conditions (against a quiet sun), and Mercury probably being impossible to find under any conditions.

B. Atmospheric Noise

The main contributing factors in atmospheric noise are variations in air mass, absorption effects due to ozone, water, and CO_2 , variations in turbidity due to dust, haze, and aerosols, and seeing effects, including scintillation noise. Since most of these factors have a more severe effect upon the total light transmitted than upon the ratio of transmission at different frequencies, the use of a color signature is particularly helpful in keeping these effects to a minimum. In the following, we will consider particularly the differential color effects of these various noise sources.

1. *Air mass effects:* In conventional photometric work, extinction coefficients

are computed from observations of standard stars, and are used to correct the observed magnitude of a star for variations of air mass with altitude. Since the extinction effect is frequently dependent, a color extinction coefficient should be based on stars which are as similar to the observed star in spectral characteristics as possible, and very narrow bands would have to be used, if a program star is to be corrected for extinction by comparison with other stars. In the program which is proposed here, extinction coefficients should be computed for the particular stars being observed, rather than being derived from comparison stars, to guarantee the proper spectral composition. An introductory discussion of extinction computations can be found in Hardie (1962). The linear equation used for air mass corrections is

$$C_0 = C - k_c X,$$

where C_0 is the observed color index, C is the color index for zero air mass, and X is the air mass (approximately equal to $\sec z$). The accuracy of k_c , the color extinction coefficient, is typically about 3%, and over a four-minute period of observation (sufficient to observe an initial or terminal eclipse transient) the air mass variation for a star within 30° of the zenith is only about 0.0002. Based on these figures, for a star at the zenith, observed from a site at the equator, at $\lambda 4000$ and $\lambda 7000$, giving $k_c = 0.24$, the color change due to air mass over four minutes would be

$$\Delta C = \Delta(R - B) = 0.0002 \times 0.24$$

$$= 5 \times 10^{-5},$$

and the error (assuming 3% accuracy in k_c) would be only 1.5×10^{-6} . At higher latitudes, where zenith distance changes more slowly, conditions are considerably more favorable. For example, for a star with $\delta = 28^\circ$ observed at Lat 36° , with $h = 0$, a four-minute change in X is about 0.0006, giving an error of about 5×10^{-6} . With $h = 2$, ΔX would be increased to 0.002, and the error correspondingly increased to 1.5×10^{-5} . This demonstrates the advisability of restricting observations

to stars near the zenith, but also demonstrates that even without further correction conventional extinction measurements could yield highly accurate estimates of these effects. By taking account of barometric pressure changes and temperature and humidity measurements, and by measuring k_c for the particular star being observed, additional improvements should be possible. These air mass effects would, of course, be swamped by variations in transparency, which we hope to reduce by pooling data from a number of telescopes at different sites.

2. *Absorption bands*: The main sources of atmospheric absorption are due to ozone, oxygen (in the infrared), water, and CO_2 . Of these, the most variable is likely to be water absorption. The oxygen effects are relatively stable. The effects of varying water content can be minimized, first of all, by selection of a high-altitude desert site (although this may increase dust and cirrus effects), and secondly by restricting observations to spectral ranges in which these bands are unimportant. In the range from 3000 to 6000 Å, water absorption is negligible. From 6000 to 7000 it is on the order of 1% at sea level, and from 7000 to 9000 increases to about 5%. Over 9000 Å water absorption would limit observations to carefully selected bands in the infrared. Since there are, in any event, serious technical difficulties in obtaining accurate measurements in this range, it seems desirable to avoid it entirely, and to consider filtering out the water absorption bands in the near infrared if this range is to be used.

To avoid ozone effects, the V band should be kept above 4000 Å, and the red band should be kept above the Chappuis bands (~ 6000 Å).

3. *Turbidity effects*: While the preceding sources of variation permit a number of systematic treatments to minimize their effect, turbidity variation, due to dust, haze, etc., is much harder to deal with. At sites close to industrial centers or cities, these effects are likely to be prohibitive. Even at a well-chosen desert site, variation can be as much as 1% over a short time period (i.e., an hour or less). One means of

estimating turbidity effects is by means of a set of comparison stars, chosen as close to the observed star as possible. Another means of correcting for the effect is by means of a statistical regression equation, which assumes that all residual variation after compensating for air mass, humidity and other systematic effects is due to turbidity changes. In this case, residual change in blue light (after applying systematic corrections) can be correlated with the changes in red, and ΔR predicted from the observed ΔB . A multiple regression equation, taking account of meteorological as well as photometric variables and including data on comparison stars would be computed for each program star. This is the method which we propose to use.

Since increases in dust or haze which result in a decrease in blue light transmitted through the atmosphere result in a *smaller* change in red light, the predicted differential effect due to turbidity, $\Delta B - \Delta R$, will have a sign opposite to that observed for an eclipse. Molecular scattering is approximately proportional to λ^{-4} ; for aerosol particles of radius less than the wavelength of the light, scattering remains an inverse function of wavelength (Van de Hulst, 1949; Robinson, 1968). For larger particles, scattering shows a fluctuating dependence on wavelength, gradually becoming independent of wavelength when the radius becomes sufficiently large. Haze droplets in a dry, clear atmosphere have diameters of about $1\ \mu$, so that scattering can be expected to decrease with increasing wavelength up to about $5000\ \text{\AA}$. For typical dust conditions, the extinction coefficient is proportional to $\lambda^{-1.3}$. With dust or water droplets of greater diameter, the scattering will continue to decrease at still longer wavelengths. Thus, if blue measurements are made under $5000\ \text{\AA}$ and red measurements above $5000\ \text{\AA}$, the predicted change in red should always be less than the observed change in blue, for any variations in atmospheric turbidity.

The actual extent of turbidity variation should be investigated for a number of sites. Kondratyef (1969) offers several illustrations of atmospheric transparency variation over a 21-minute period on

different days. On the most stable of three days, variation remained within a 10% range, while on another day a 50% variation was observed within one minute. Such sudden changes are particularly likely to occur on windy days, which suggests that observations should probably be limited to calm periods, as well as to dry, high-altitude sites with good ground cover, well removed from cities or sources of industrial pollution.

4. *Seeing and scintillation noise*: Seeing effects, due to atmospheric turbulence and the movement of cells of varying density in front of a telescope, are known to be approximately frequency-independent for large apertures (Young, 1967, 1969a). This means that if measurements are made simultaneously in red and blue light, observed through the same optical system, so that both channels are subject to the same scintillation noise, the red/blue ratio should be unaffected, even though there may be substantial variation in the absolute measurement on each channel. Since we propose to use variations in blue measurements to predict correlated variations in red, due to turbidity, it is still important to distinguish turbidity variation from scintillation noise, however (as discussed in Section V).

Young (1967) has shown that for observations near the zenith low-frequency scintillation noise increases with the $3/2$ power of the air mass, and with the $-4/3$ power of the telescope aperture. He estimates the rms error for a 10-second observation under typical conditions, with unit air mass, to be 0.0010 mag for a 16-in. telescope, 0.0006 for a 36-in., and 0.0004 for 60-in., at sea level. At 2000 meters altitude, the corresponding errors would be reduced to 0.0008, 0.0005, and 0.0003, respectively. All of these errors are approximately tripled at two air masses. For telescopes of 16 in. or larger, scintillation noise can be expected to exceed photon noise for all stars brighter than the 6th magnitude. Thus if we require scintillation noise to be less than 10^{-4} mag, with a 36-in. telescope at 2000 m, with unit air mass, we would require an observation of 240 seconds, which is perfectly acceptable for the

detection of the transients which we hope to observe. If we required an accuracy of 10^{-5} , however, 24 000 seconds would be required, or else a great increase in telescope aperture, either of which would be prohibitive. An alternative means of reducing noise, which is proposed in the following section, is to employ a battery of smaller telescopes, widely separated so that they are observing through statistically independent pencils of atmosphere. In this case, for n telescopes, all observing the same event, the noise due to all atmospheric effects (turbidity as well as scintillation) is reduced by the square root of n . Since this also provides a coincidence system which guards against electronic artifacts and instrumental errors, it appears to be preferable to simply increasing the aperture of a single telescope.

High frequency scintillation noise is more strongly aperture dependent (varying approximately with the inverse square of the aperture), and is markedly affected by a central stop or spider in the telescope. Nearly all scintillation noise occurs at a frequency under 100 Hz. Thus essentially simultaneous observation in two colors is provided, with a single channel, by chopping at a frequency greater than 100 Hz. This may prove to be preferable to simultaneous counting in two channels, not only for economy, but also because differential drift of electronic components for the two channels is thereby avoided.

C. Instrumental Noise

1. *Shot noise and thermal variation:* The statistics of the arriving photons give an irreducible minimum noise level of $\sqrt{n}$, where n is the expected number of photons entering the telescope aperture. To minimize this effect, it is desirable to employ a photocathode with maximum quantum efficiency in each of the bands observed, to keep the observations in the most energetic spectral range of the star, and to use bands which are as wide as possible. Table III shows the power distribution and corresponding photon flux per square meter at sea level for a star of magnitude 3.2 having a spectrum identical to that of the sun. It

also gives the quantum efficiency of an S-20 photocathode over this range, and the expected counting rates for each 1000 Å band. By combining these data with the expected limb darkening coefficients for the different spectral bands, we can predict the signal/noise ratio to be expected for different band choices. This is shown in Table IV.

This table shows $\Delta R - \Delta V$ as the "signal measure." It will be shown in the following section that the signal/noise ratio can be improved by making use of the difference $\Delta R_{\text{obs}} - \Delta R_{\text{pred}}$, rather than $\Delta R - \Delta V$, where ΔR_{obs} is the observed change in the red band, and ΔR_{pred} is the predicted change, based on the observed value of ΔV and adjusted for turbidity effects. Whichever method is used, however, it is important to maximize the signal/noise level shown in Table IV.

This analysis suggests that the best choice of bands, if we disregard atmospheric effects, would be a V-band with an upper cutoff at about 5000 Å, and an R-band with a lower cutoff at about 6000 Å. To avoid ozone effects, however, it would be desirable to raise this cutoff towards 7000 Å. This must be balanced against the desirability of cutting off the infrared range prior to the main water absorption bands. In addition, the bands actually employed may have to be much narrower than those suggested in Table IV, to escape from high-order effects in extinction, particularly if strong reliance is placed on comparison stars.

The strong dependence of photomultiplier and filter color response on temperature (Young, 1963) can also be better defined with a narrower bandwidth. The dependence of the photomultiplier response on temperature may be as great as 1%/°C. With an S-20 surface at dry ice temperatures, cooling tends to raise the sensitivity in the blue and reduce it in the red. For bands centered at 4500 and 7000 Å, temperature dependence seems to be quite similar, and variations in color ratio (for selected tubes) might be less than 0.1%/°C. Nonetheless, it is clearly important to control temperature to a small fraction of a degree at the photomultiplier. This may

TABLE III

TYPICAL STELLAR POWER SPECTRUM AND COUNTING RATES FOR AN S-20 PHOTOCATHODE

λ	Solar power w/m ²	Photon flux/m ² star $m = 3.2$	S-20 ^a quantum efficiency (%)	Expected counts per 1 m ²	Limb darkening at $r = 0.98$
3-4000	50	8×10^7	18	1.4×10^7	30%
4-5000	130	29×10^7	16.5	4.8×10^7	32
5-6000	140	39×10^7	10	3.9×10^7	42
6-7000	130	42×10^7	4	1.7×10^7	55
7-8000	100	38×10^7	1	0.4×10^7	59
8-9000	80	34×10^7	0	—	64
9-10 000	50	24×10^7	0	—	66
10-11 000	47	25×10^7	0	—	67
11-12000	37	21×10^7	0	—	69

^aQuantum efficiency based on ITT data.

TABLE IV

COMPUTATION OF SIGNAL/NOISE RATIOS FOR A 1% ECLIPSE, FOR $m = 3.2$ STAR WITH DIFFERENT BAND COMBINATIONS, USING S-20 PHOTOCATHODE AND 1 m² TELESCOPE, FOR 1 SEC (SHOT-NOISE ONLY)

V-band	R-band	Counts/sec/m ²		Δn for 1% eclipse		Percentage change		Signal	Signal/
		V	R	V	R	ΔV	ΔR	$\Delta R \cdot \Delta V$	noise
3-4000	7-8000	1.4×10^7	0.4×10^7	4.2×10^4	2.4×10^4	0.30	0.59	0.0029	5.1
3-4000	6-8000	1.4×10^7	2.1×10^7	4.2×10^4	11.7×10^4	0.30	0.56	0.0026	7.4
3-4000	5-8000	1.4×10^7	6.0×10^7	4.2×10^4	28.1×10^4	0.30	0.47	0.0017	5.7
3-5000	7-8000	6.2×10^7	0.4×10^7	19.6×10^4	2.4×10^4	0.32	0.59	0.0027	5.2
3-5000	6-8000	6.2×10^7	2.1×10^7	19.6×10^4	11.7×10^4	0.32	0.56	0.0024	9.6
3-5000	5-8000	6.2×10^7	6.0×10^7	19.6×10^4	28.1×10^4	0.32	0.47	0.0015	8.4
3-5500	5500-8000	8.2×10^7	4.1×10^7	27.8×10^4	19.9×10^4	0.34	0.49	0.0015	7.9
3-5500	6-8000	8.2×10^7	2.1×10^7	27.8×10^4	11.7×10^4	0.34	0.56	0.0022	9.0
3-6000	7-8000	10.1×10^7	0.4×10^7	36.0×10^4	2.4×10^4	0.36	0.59	0.0023	4.5
3-6000	6-8000	10.1×10^7	2.1×10^7	36.0×10^4	11.7×10^4	0.36	0.56	0.0020	8.4

$$\text{Noise} = \sqrt{1/n_v + 1/n_r} \text{ where } n_v = V \text{ counts/sec and } n_r = R \text{ counts/sec}$$

be done in part by a well designed cooling system, and in part by measurement and correction of data for residual temperature variations.

If very narrow bands are used, then photon noise rather than atmospheric effects becomes the limiting factor in system performance, and large aperture telescopes may be required. The optimum choice of bands therefore represents a compromise between available telescope

aperture and the desirability of narrow bands to improve the accuracy of corrections for atmospheric and thermal effects. An empirical study with particular tubes, under actual observing conditions, seems to be required before an optimum band selection can be made.

2. *Integration and counting errors:* Of the two basic alternative techniques for measuring light intensity, current integration or photon counting, the photon

counting technique (or rapid A-D conversion every few milliseconds) is probably more appropriate. At high light levels, such as we assume here, Young (1969b) concludes that pulse counting may have an advantage of a factor of 2 over dc methods for detection, but dc techniques have better linearity. At low light levels, pulse counting is increasingly advantageous. For short observations of a few seconds, such as might be carried out to detect the signature of a large planet, either technique would be suitable. For detecting signal differences on the order of 10^{-4} , however, integration times of a half hour or longer may be required. In this case (and also for comparison of widely spaced measurements) the main consideration is to minimize drift due to changes in electronic components, power supplies, or thermal effects. For this reason, the pulse-counting or rapid digitization method seems desirable. To avoid the differential drift which might occur between components in the two channels of a multichannel photometer, it may be preferable to use a single-channel system, chopping between the two colors at a high enough rate to avoid differential scintillation effects.

With a pulse counting method, employing a suitable refrigeration system and pulse-height discriminator, dark noise can be reduced to a negligible level. It can be measured together with sky background and cosmic ray contributions, by switching the image dissector to observe a sky area during each cycle of observations. The use of a small effective aperture in an image dissector reduces noise from a large photocathode to a very low level, but requires a high degree of mechanical rigidity and freedom from flexure to eliminate effects of nonuniformity of the cathode surface. Moreover, an excessively small aperture would tend to lose light from a diffuse or off-axis star image. For these reasons a relatively large effective aperture is desirable. With an effective photocathode diameter of 0.1 in., an S^c-20 tube can be expected to give a dark-counting rate of 10 pulses per second, with a possible maximum of 100 per second. This

represents a negligible addition to the shot noise for a bright star, where the net counting rate may be on the order of 10^6 or 10^7 per second. With faint stars or narrow spectral bands, however, this component might become significant.

With very bright sources, an additional source of error may be the coincidence of several photons, too closely to be discriminated, or the jamming of the counter system. Counting rates of 10^7 per second can be achieved, however, with available equipment, which is quite adequate for the levels of accuracy which we require. Since we are not concerned with absolute measurement, but with measurement of slight changes in the signal, minor deviations from linearity are unimportant, as long as stability can be maintained over the period of observation. Since we propose to time-share the system for a number of stars, the remaining stars can be used as a check on the stability of the system while measurements are made on any one member of the set. The use of multiple telescopes, each with its own power supply, photomultiplier, and counting system, is a further guarantee against the acceptance of spurious signals which are due to electronic artifacts in one of the systems. Cosmic ray showers could still result in some correlated noise, but this would be included in sky background counts.

3. *Tracking errors:* The wandering of the image, whether due to seeing effects or to improper telescope guiding, can add a noise component to the signal. In part this may be due to loss of signal when the star (or a portion of the stellar image) drifts out of the effective aperture, and in part it is due to variations in sensitivity of different small areas of the photocathode. The first effect is of negligible importance in a one-channel system, but could be appreciable in a two-channel system where the image might be momentarily lost to one channel but not to the other. The effective aperture of the image dissector should be wide enough so that minor displacements of the image due to seeing will not result in a substantial loss of signal. An effect that might be more important when trying to

obtain measurements of the highest precision is the appearance and disappearance of faint background stars sufficiently close to the observed star to be located within the margin of the effective aperture. In crowded regions of the sky, this might lead to the intermittent inclusion and exclusion of additional light of a different spectral composition.

As an example, suppose we are observing a field having a density of one star of program brightness (program stars plus other bright stars) per square degree. If we assume an effective aperture of 2 arc minutes, and are seeking an accuracy of 10^{-4} (corresponding to stars 10 magnitudes dimmer than the program star) then we would expect an average of about 100 faint background stars in the field during a typical observation. (There would, of course, be even greater numbers of still dimmer stars, but their light would tend to be included in sky brightness measurements, and would be less likely to lead to sudden step functions in observed luminosity due to slight shifts in the field boundaries.) Since a guidance error of $6''$ of arc would result in a shift of about 10% in the observed area, we can estimate that such an error would result in an exchange of about 10 faint stars, or a standard deviation of about 3×10^{-4} of the observed luminosity. If the guidance error is periodic, this will tend to be smoothed out over time, and such errors can be further reduced by the use of multiple telescopes. Nonetheless, it is clearly important to keep guidance errors to a minimum. In an image dissector under computer control, such tracking would be done electronically rather than by mechanical motions of the telescope. With a rigid system and continuous feedback control, we may assume that residual tracking errors can be held within 2 arc seconds, which should hold this noise source to a level less than 10^{-4} in total luminosity, and appreciably less in color index.

The effects of variations in photocathode sensitivity, due to blemishes, "hot spots," or other variations of the surface, must also be reduced by maintaining a high degree of instrumental stability. To the extent that

these variations are uniform in the red and blue bands, they will not affect differential measurements, but in general they are color specific, and will therefore add a noise component which must be assessed for the particular tube employed. As with other types of electronic noise, the best assurance that an observed effect is not due to an artifact of this type is to employ a multiple coincidence technique, with a number of independent telescopes.

Slight variations in filter density can also cause signal variation (and apparent color variation) with a wandering image. To minimize this effect, it is desirable to avoid placing filters close to the focal plane, so that light from each star is distributed over a substantial part of the filter surface and local variations are averaged out.

V. METHODS OF DATA COLLECTION AND ANALYSIS

We now turn to the technique to be used for data collection, with two particular questions to be answered: (1) What technique can be used to detect eclipse signals most efficiently, and with minimum effect of atmospheric noise? (2) What search strategy should be used to guarantee a maximum rate of discovery of planetary systems? In the following, we shall consider each of these problems in turn, and propose methods which seem appropriate in each case. It will be seen, however, that the choice among alternative methods really requires data which are not available at this time, concerning short-term atmospheric stability and differential effects over small angular separations, at the level of accuracy and in the particular spectral bands at which our measurements are to be made. For this reason, a preliminary program to examine alternative data processing techniques at various sites would seem to be desirable before a final system design is adopted.

A. *The McCullough Study*

An interesting baseline for such a program is offered by James McCullough's (1964) study of short-period variable

stars. Although he was measuring V magnitudes, rather than color indices, the study was in other respects similar to what we are proposing. Using a 16-in. telescope and a 1P21 photomultiplier with a V filter, he observed 483 bright stars near the zenith, looking for short-term changes in magnitude. Although the photomultiplier was not cooled, a linear dependence on temperature was found by means of which temperature corrections were made. Integration times were short, and measurements were referred to the averages of sets of comparison stars, after correction for atmospheric extinction. Residual errors were reported to have a standard deviation of 0.005 mag. Actually, however, this is a misleadingly high estimate of the experimental error, since all deviations were computed from the most extreme observation on each night, rather than from the mean. Thus what is reported as σ is at least double the value that would have been obtained from a conventional error analysis. Since the range of observations tends to increase with the number, the function reported by McCullough as " σ " is actually a divergent rather than a convergent function of sample size. We might safely infer, therefore, that McCullough actually achieved a standard deviation <0.002 mag in these observations. This is particularly impressive in view of the relatively simple analytic procedures and equipment used, the use of weather reports and subsequently of air temperatures for data correction instead of thermal control of his photomultiplier, and the use of a high turbidity, low altitude observing site in Connecticut. Had color ratios been measured, it is likely that an error <0.001 mag would have been observed. It is also noteworthy that differences observed over intervals of 6 days or longer seemed to be stable within these error limits.

We now turn to an examination of techniques which might be employed to further reduce this error, for purposes of the present investigation.

B. *The Use of Predicted Differences*

In the discussion of colorimetric signatures in Section III, and in the signal/noise

computations in Table IV, it was assumed that the ratio of red to blue luminosity (or, equivalently, the difference in magnitudes) would be computed directly by taking the ratio of photon counts in the red and blue bands. For space-based observations, this would indeed be the appropriate procedure. For ground-based observations, however, where spurious signals might be generated by atmospheric variation, an alternative method appears to be more attractive, both as a means of strengthening the apparent effect and as a means of reducing the atmospheric noise component. This method is to use the observed change in the blue (V) band to predict the change that would be expected in the red (R) band, on the hypothesis that the observed change is due entirely to turbidity changes, or other atmospheric perturbations. Since, in the case of an eclipse, the predicted change in R will be much less than the observed change in R , the use of $\Delta R_{\text{obs}} - \Delta R_{\text{pred}}$ as a measure of effect can be expected to enhance the signal/noise ratio considerably.

As noted in the previous section, the extinction coefficient due to dust, haze, and other aerosol components shows a wavelength dependence of the form $\lambda^{-\alpha}$, where α may vary between 1 and 2 depending upon time and local conditions. A value which has been found to hold well for standard conditions (Robinson, 1968) is $\alpha = 1.3$. Figure 4 shows the function $\Delta R_{\text{obs}} - \Delta R_{\text{pred}}$ for a 1% eclipse, where the blue and red bands are 4500 and 7000 Å, respectively, for different values of α . Note that in all cases, the observed-predicted function is of substantially greater magnitude than the direct method (shown in the broken curve) of measuring the color ratio per se. Because the atmospheric effect is stronger in the blue than the red ($\alpha > 0$), while the eclipse effect is stronger in the red than the blue during its initial and terminal transients ($\alpha < 0$), the sign of the difference $\Delta V - \Delta R$ tends to be opposite in the case of a turbidity change than in the transient phase of a stellar eclipse. If the signal is integrated for the first tenth of the eclipse duration (from $R = 1.0$ to 0.8) then the integrated signal using the prediction method is more than four times as large as

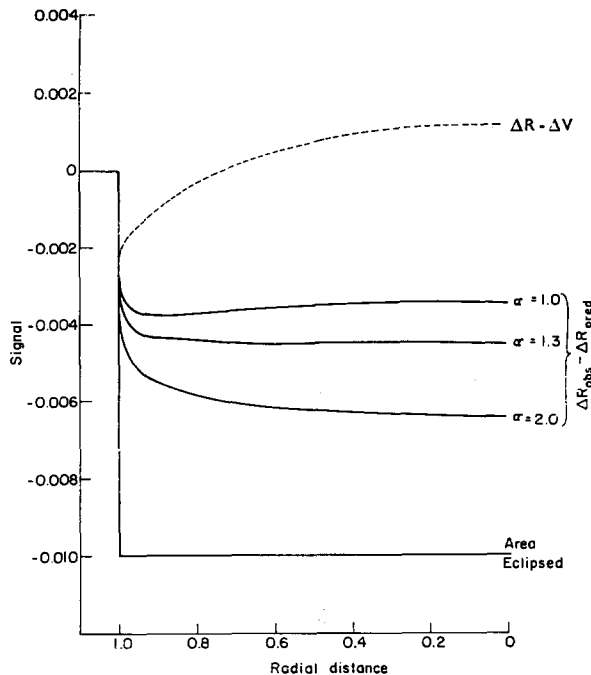


FIG. 4. Signature curves for eclipse of 1% of solar area, using observed-predicted change as measure of eclipse. $\lambda_V = 4500 \text{ \AA}$; $\lambda_R = 7000 \text{ \AA}$.

the integrated signal using the direct ratio method, with $\alpha = 1.3$ (the middle curve in Fig. 4).

This does not necessarily mean that the signal/noise ratio can be enhanced four times by this means, as that will depend upon the errors in predicting ΔR . This prediction depends upon the accuracy of our measurements of α , and the effect upon the prediction equation of other noise sources, such as scintillation noise. We must therefore consider the means by which ΔR can be predicted most accurately for a stable star, where all variation is assumed to be due to atmospheric or instrumental noise sources.

In this analysis, we assume that shot noise and instrumental errors can be neglected, the principal sources of variation being changes in air mass, changes in turbidity, and scintillation effects. As a first approach, let us assume that measurements are made simultaneously in both V and R bands, for a number of close stars of

similar spectral types (which in practice are hard to find!). The primary data are four measurements for each star:

$$V(t_1), V(t_2), R(t_1), \text{ and } R(t_2),$$

where V and R refer to logarithms of photon counts, and t_1 and t_2 are two successive periods of observation. Including all main sources of variation, we have:

$$V(t_2) = V(t_1) + k_v \Delta X + \Delta T + \epsilon_v + \Delta N, \quad (6)$$

$$R(t_2) = R(t_1) + k_r \Delta X + k_t \Delta T + \epsilon_r + \Delta N \quad (k_t < 1), \quad (7)$$

where ΔX represents the change in air mass (corrected for variations in barometric pressure, zenith distance of the observed star, etc.), ΔT = change in atmospheric turbidity measured as a change of magnitude in the V band, ΔN = scintillation noise, in magnitudes, ϵ_v = eclipse effect or

other stellar variability in V band, and ϵ_r = stellar variation in R band. We assume that ϵ_v and ϵ_r are zero, for present purposes, and that scintillation noise is independent of wavelength. To predict ΔR as a function of ΔV , we obtain from (6) and (7)

$$\begin{aligned}\Delta R &= R(t_2) - R(t_1) \\ &= \Delta V + (k_r - k_v) \Delta X + (k_t - 1) \Delta T \\ &= \Delta V + k_c \Delta X + (k_t - 1) \Delta T \quad (8)\end{aligned}$$

where k_c = color extinction coefficient (simplified for this example by neglecting higher-order terms); k_t = ratio of turbidity effect in V and R bands = $(\lambda_v/\lambda_r)^\alpha$. Note that the scintillation noise, having the same proportional effect in both bands, has vanished from this equation (which disregards some residual second-order effects). The accuracy of prediction then depends upon the accuracy with which we can estimate k_c , k_t , ΔX , and ΔT . The air mass effect is relatively systematic, and the main uncertainty is in k_t and ΔT , which is more likely to depend on transient conditions. An estimate of k_t can be made directly from Eqs. (6) and (7) by observing a number of standard stars for a long enough period so that ΔN is negligible, and performing a least-squares fit for k_t . In that case, Eq. (8) can be solved for ΔT for each comparison star, and a suitably weighted mean $E\Delta T$ can be used with Eq. (8) to predict ΔR for the "target star." This procedure is one which would work well if turbidity changes occur uniformly over the set of comparison stars. It is subject to appreciable error, however, if turbidity changes occur differentially for the target star and comparison stars, or if turbidity effects are large relative to scintillation noise. If that should be true—as is likely to be the case if wisps of dust or haze move across different comparison stars at different times—then a different method is recommended, which accepts some residual scintillation noise, but avoids the use of comparison stars to estimate ΔT .

In order to minimize differential effects between comparison stars, we can lump the variation due to turbidity and scintil-

lation noise, letting $D = \Delta T + \Delta N$. Then from Eqs. (6) and (7) we obtain:

$$\Delta V - k_v \Delta X - D = 0, \quad (9)$$

$$\Delta R - k_r \Delta X - k_d D = 0, \quad (10)$$

where $k_d = k_t$ if $\Delta N = 0$. From these equations we obtain

$$\Delta R = k_d \Delta V + (k_r - k_d k_v) \Delta X \quad (11)$$

which can be used to predict ΔR from measurements on the target star alone, provided the coefficients k_r , k_v , and k_d are known. Since k_d is more likely to be stable from star to star than ΔT , it seems most appropriate to measure this coefficient from a set of close comparison stars, applying the mean value in Eq. (11) to predict ΔR . The effect of including the scintillation noise, ΔN , as part of the quantity D is to introduce an error in the estimate of ΔR which is equal to

$$|(k_t - 1) \Delta N| < |\Delta N|.$$

For the case of $\alpha = 1.3$, $k_t = 0.56$ and the error will be about $0.44\Delta N$, for the case illustrated in Fig. 4. Thus the effect of this treatment is to reduce the combined effects of turbidity variation and scintillation noise to an over-all level of about 0.44 times the scintillation noise level. The value of k_d should, of course, be continually reevaluated from a set of close comparison stars of "similar" spectral types, to allow for changes in the size distribution and types of aerosol particles in the atmosphere. In performing these calculations, a least squares fit which includes corrections for color index is probably desirable, and the narrower the spectral bands employed, the more accurately k_d can be determined.

The signal/noise ratio which can be expected from the application of Eq. (11) to the signal curves shown in Fig. 4, based on typical scintillation figures given by Young (1967) is shown in Table V. This assumes that all observations are made on stars of sufficient brightness so that scintillation, rather than shot noise, is the limiting effect. The signal/noise ratio, therefore, is independent of the magnitude of the star as long as this condition holds (unlike the shot-noise calculations given in Table IV). Note also, that the signal/

TABLE V

SIGNAL/NOISE RATIO FOR 10-SECOND OBSERVATIONS OF A 1% ECLIPSE AT $R = 0.8$
(BASED ON CURVES IN FIG 4) ASSUMING TYPICAL SCINTILLATION NOISE AT AN
ALTITUDE OF 2000 m

Air Mass	16-in. telescope			36-in. telescope		
	$\alpha = 1.0$ $k_t = 0.64$	$\alpha = 1.3$ $k_t = 0.56$	$\alpha = 2.0$ $k_t = 0.41$	$\alpha = 1.0$ $k_t = 0.64$	$\alpha = 1.3$ $k_t = 0.56$	$\alpha = 2.0$ $k_t = 0.41$
$X = 1.0$	12.9	12.6	12.3	20.5	19.5	19.8
$X = 2.0$	4.7	4.6	4.5	8.0	7.5	7.6

noise ratio is nearly independent of α , since the increased noise level with larger values of α is offset by the increased difference between observed and predicted variations. The signal/noise ratio is similarly insensitive to the exact choice of red and blue wavelengths, even though the signal itself (shown in Fig. 4) would be increased by further separation of the red and blue bands.

It is important to compare these results with those which might be obtained by a direct observation of luminosity changes. In the case of eclipses by large planets, such changes are well above the noise level which might be expected from stellar noise, and observations will be limited chiefly by atmospheric effects. First of all, note that the signal magnitude will typically be somewhat less than double that obtained from the method described above (depending upon the magnitude of α and the time at which the eclipse is observed). In principle, there could be a gain of about a factor of 3 in counting efficiency, by eliminating the filters and using continuous observation rather than chopping between bands. Thus the signal/noise ratio for shot noise effects would be improved by a factor of about 5 from these combined effects. On the other hand (if we disregard the extinction problem, which becomes intractable without filters), scintillation noise remains the limiting factor, and this will be roughly twice as great as in the colorimetric method, just offsetting the gain in signal size. Thus, if we disregard probable turbidity effects, the two methods (for large eclipses) would be closely competitive, with no clear advantage for one or the other. The real difficulty comes from

our inability to devise a technique for controlling turbidity noise using total luminosity measurements which is comparable to that described above. While we could limit such effects by star-to-star comparison (as in the McCullough study) using a number of close comparison stars, we have already seen that this is likely to be less than satisfactory due to the inhomogeneity of atmospheric dust and haze content. Finally, the relative sensitivity of total luminosity measurements to equipment variations, vignetting, and other effects, makes it more likely that artifacts will occur through this procedure than through a colorimetric method. The use of total luminosity measurements should probably be restricted, therefore, to a confirmatory technique, or possibly as a rapid survey technique with more definitive measurements relying on the colorimetric procedure when a suspected signal has been detected.

C. Search Strategies

Two main alternatives may be considered. (1) We may survey a relatively small number of bright stars of appropriate spectral types, with observing times long enough to detect any eclipses which are likely to be above the stellar noise level (about 0.0001). (2) We may survey a much greater number of stars, reaching out to a fainter magnitude, but with observation times reduced so that only giant planets (say eclipses >0.005) would be detected. Assuming that the orbits of such planets (which would include Jupiter and Saturn in the case of the Solar System) are likely to include a number of lesser planets at

distances closer to their primary, we could then commit one telescope to a continuing search of these already identified systems with much longer integration times, and with a high probability of detecting the inner planets in a reasonable time. We must ask which of these procedures is likely to yield a higher rate of detection with a given system, or, alternatively, what is the relative size and complexity of the system required to achieve an expected rate of, say, one discovery per year employing each of these strategies?

We have already seen, in Section II, that to detect one of our inner planets would require about 3000 observations, each made over a 6-hour period. Since it is unlikely that all of the inner planets would be of detectable size, a more reasonable number to aim for would be about 10 000 observations per year (using a time-sharing system) to have an expected detection rate of one system per year. A rough estimate based on spectral types listed in the *Atlas Coeli* (Bečvář, 1959) suggests that for any one observing station, excluding binaries and giant stars, there are about 500 suitable stars of late spectral types down to mag 6.25. This magnitude seems to be a reasonable cutoff for a small telescope if we are not to be limited by shot-noise. This is a sufficient number of stars so that we would expect 5 systems to have orbital planes so oriented that all planets out to the orbit of Mars would be detectable. To accumulate 10 000 observations per year, it would be necessary to observe about 30 stars each night. If this is done with 16-in. telescopes, and observations are limited to stars near the zenith, then (based on Table IV, and allowing a factor of 2 for chopping between two filters) we would need 30 telescopes operating in parallel, each tracking a different star and integrating for periods of 1 hour per observation to give a S/N ratio of 1.67 from an eclipse of 10^{-4} . With this ratio, one observation in 20 would be above the S/N level, and would call for more intensive study, the entire battery being diverted from other assignments long enough to verify whether an effect of interest has been detected. Alternatively, all telescopes could observe the same stars,

in which case observations would be limited to 2 minutes each. This has the advantage that a suspected signal could be checked out immediately by verifying its occurrence in a majority of the telescopes, and by immediately extending the observation time to obtain additional data. In either case, it is assumed that the entire battery of telescopes would be computer controlled and separated widely enough so that high frequency atmospheric noise effects would be independent. This system has the advantage that a relatively small computer could handle all data processing, since the data rate is relatively slow, and the storage requirements are minimal. On the other hand, the number of telescopes is far in excess of the number required by the alternative strategy.

If we consider the second strategy, we find that for the same cutoff probability for taking a second look at a given signal ($P = 0.05$, corresponding to an S/N ratio of about 1.7) a single 16-in. telescope would require only about 1.2 seconds per observation. Thus, with a 1 hour cycle of observations, each telescope in a battery could observe about 3000 stars, and 3 telescopes could observe 9000 stars each night. (Setting time is neglected in this estimate, since most resets are to be done electronically within the field of the image dissector, rather than by moving the telescope.) From the data in Table I, we find that to obtain an expected rate of 1 discovery per year of planets in the orbits of Jupiter or Saturn we must monitor about 4000 stars. Thus 9000 stars would be a sufficient sample to give a high probability of discovery, assuming the solar system as a model. (The dark companion of 61 Cygni is actually closer to its primary than Jupiter, orbiting at a mean distance of about 2 A.U., in a highly eccentric orbit, as well as being much larger; also both planets hypothesized for Barnard's star are closer than Jupiter. Thus the solar system model may be conservative as a basis of estimate.) This suggests that a preliminary survey of a large number of stars for large planets is the preferable alternative, requiring only two or three telescopes and staying well above the probable noise limits of the

technique. In order to reach this number of stars, however, we should be prepared to extend the survey to approximately magnitude 8.0. The proportion of useful stars should be expected to increase as the magnitude limit becomes fainter, due to the increasing proportion of red dwarfs of late K and M types. In order to continue to obtain adequate data rates down to m 8.0, it is probably desirable to increase the telescope apertures to 20 in. or 24 in. (or possibly more if narrow spectral bands are required). In this case, a battery of three telescopes operating in parallel for verification purposes should be able to verify the occurrence of any eclipses down to 10^{-4} , in a follow-up program.

The main disadvantage of this strategy is the relatively large storage capacity required in the computing system, to handle the large number of stars. Since for each star it is necessary to store two coordinates, a previous color reading, and probably several coefficients for use in the prediction equations, a storage capacity of 50 000 to 100 000 words seems to be needed. Immediate access to all of this is not required, however, so that a disc or tape storage system with a relatively small high-speed memory sufficient to handle the set of stars in the field at one time for all telescopes would be sufficient.

An intermediate system, searching for eclipses down to 0.001, is also attractive in some ways, as a compromise. Such a system could survey 100 stars per hour per telescope, with about 40 seconds per observation, and a battery of three telescopes could make about 10^5 observations per year. Thus if we assume that there are likely to be an appreciable number of systems having inner planets 3 times the diameter of the Earth or larger, this system would be expected to detect them. The expected discovery rate for planets in the Jupiter orbit or further in this case would be reduced to about one discovery every ten years.

The final choice among these alternatives should depend upon results of a preliminary study with a single small telescope and computer, to evaluate alternative instrumentation and data processing techniques.

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